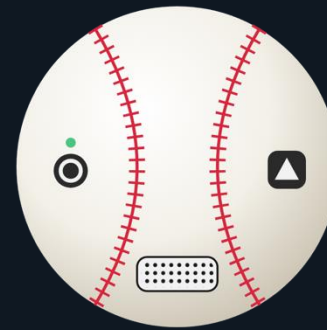


Squeeze the Beat

Research and design notes for a squeeze alarm made for
blind and low-vision sleepers



How blind people set an alarm today

88%

They ask a voice assistant

In a survey of 145 blind and visually impaired people, 88% had used Siri, Alexa or Google Assistant. Alarms and timers were among the top uses.

Problem: it mishears in noisy rooms and gets accents wrong.

Swipe

They swipe a time picker

With VoiceOver on iPhone, you swipe right to move between hour, minute and AM/PM, then swipe up or down to change each one. Every value is read out.

Familiar, but slow for big changes.

Talk + Set

They press a talking clock

Talking clocks have a big Talk button that says the time and a Set button that steps through hour and minute. Some add a vibrating pad.

Simple and tactile, but no music or phone setup.

What this means for Squeeze the Beat



Offer three ways to set the time

Voice is fastest, the swipe wheels are familiar, and the ball works without a phone.



Always read the result back

Voice can mishear, so the ball confirms: "Alarm set for 6:30 AM. That's in 8 hours."



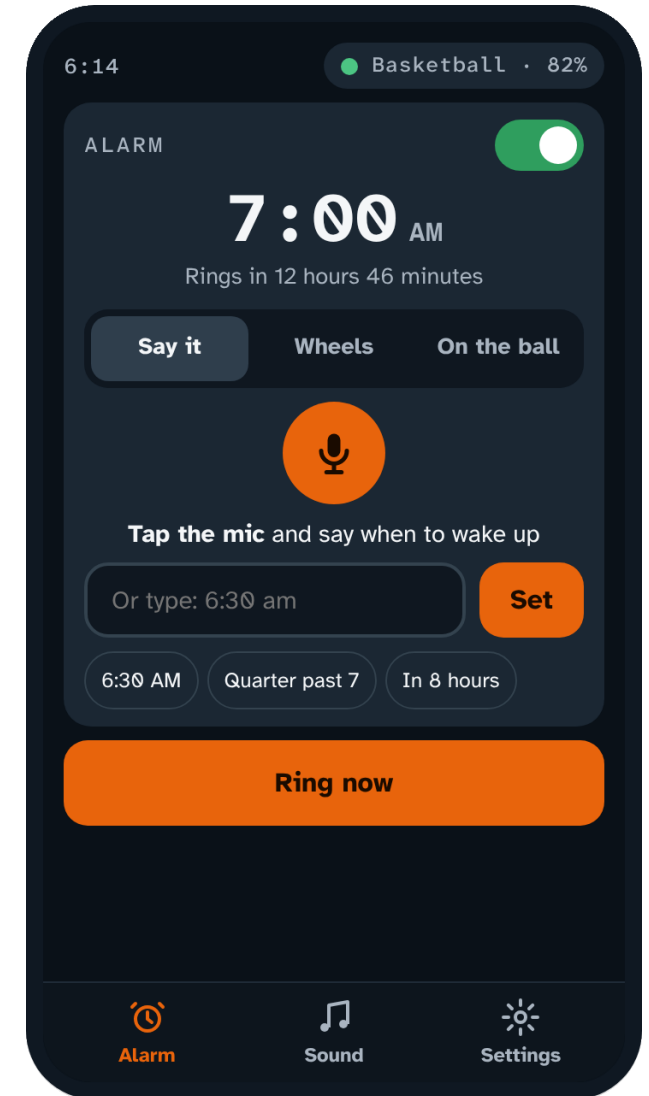
Use standard controls

Plain adjustable controls already work with the VoiceOver and TalkBack gestures people know.

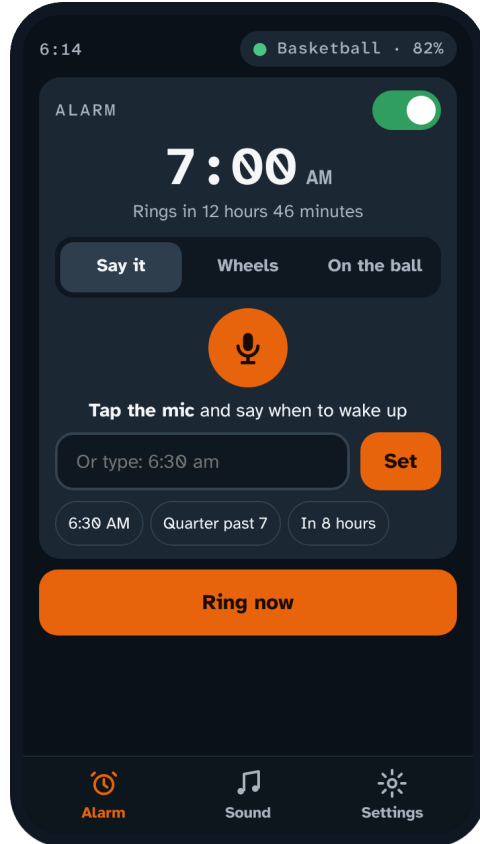


Speak every step on the ball

Each squeeze says the new value out loud, like a talking clock.

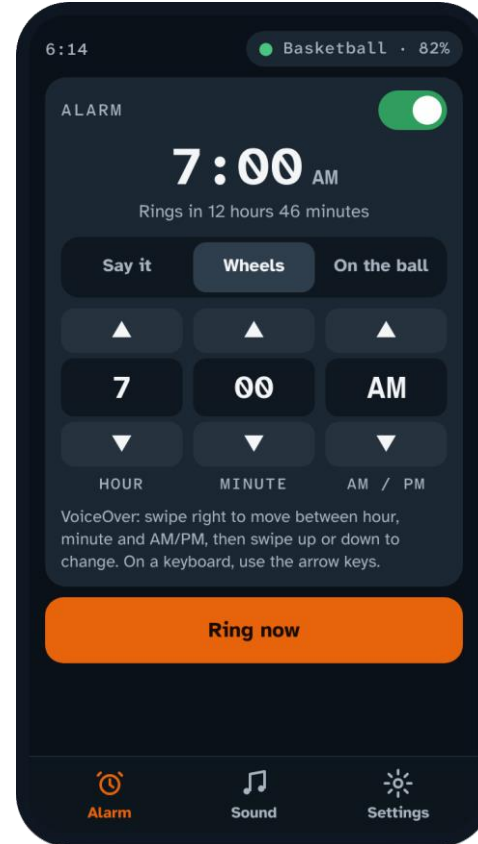


Three ways to set the alarm time



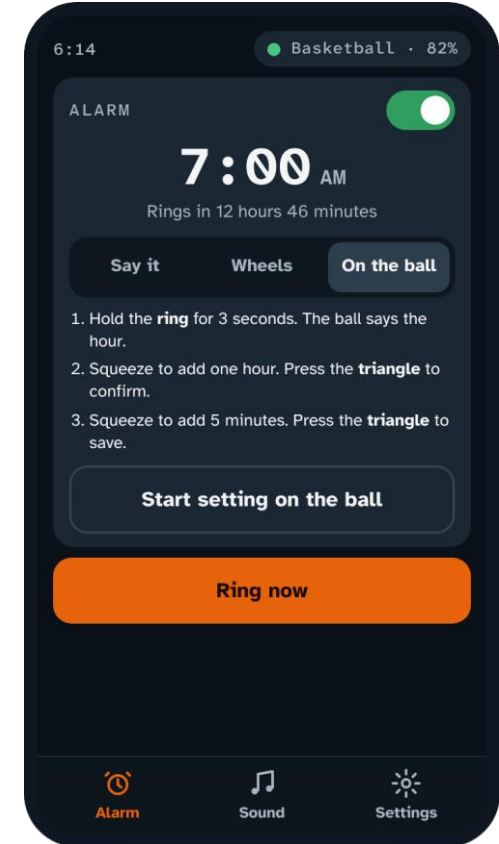
Say it

Tap the mic and say "Wake me at 6:30".
Typing works too.



Wheels

Hour, minute and AM/PM. Swipe up or down with VoiceOver.



On the ball

Hold the ring, squeeze to change, press the triangle to save.

A magnetic holder keeps the ball in place

It can't roll away

A ball that rolls under the bed is hard to find without sight.

Same spot every night

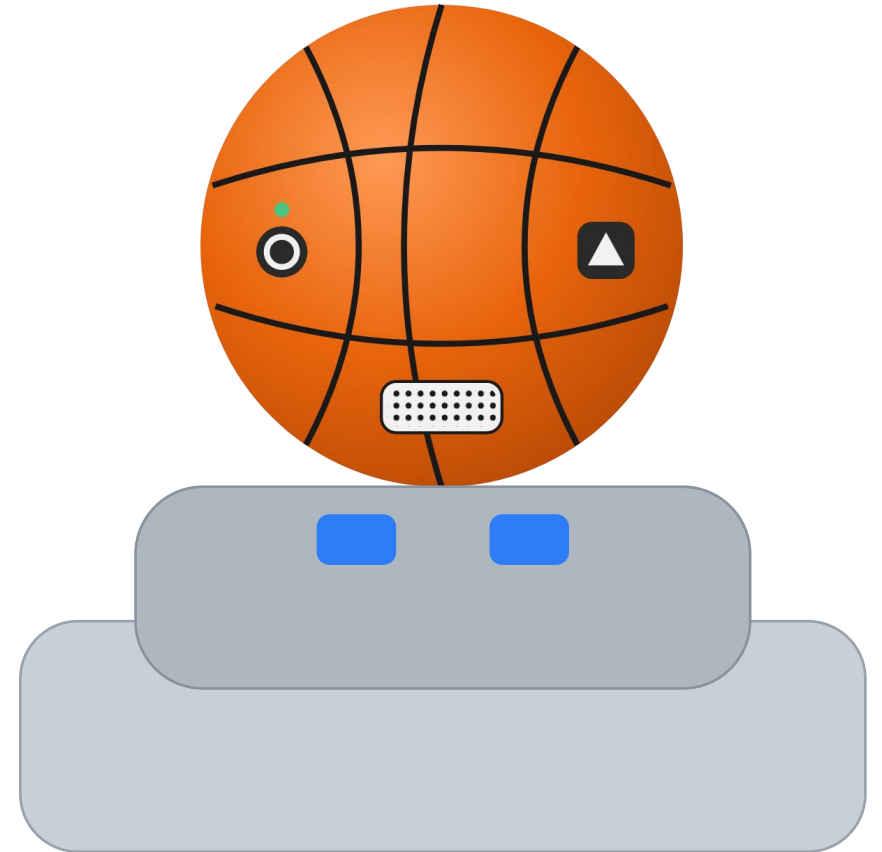
The holder stays on the nightstand, so the hand goes straight to it.

Always faces the same way

The ring, triangle and speaker are in the same place every time.

Two magnets, not one

With one magnet the ball can spin. Two side by side lock which way it faces.



Two magnets in the holder, two steel washers in the ball

The holder is also the charger

Charges whenever it sits there

Like a magnetic phone charger. No cable to find in the dark.

Magnets line up the contacts

The same two magnets that stop rolling also put the charging contacts in place.

A simple nightly habit

On the holder at night, off in the morning. Charging happens without thinking.

The battery level is spoken, not shown

A blind user can't read "82%", so the ball and the app say it.

Put on the holder

Chime, then "Charging, 82 percent."

Battery low at night

"Battery low. Put me back on the holder."

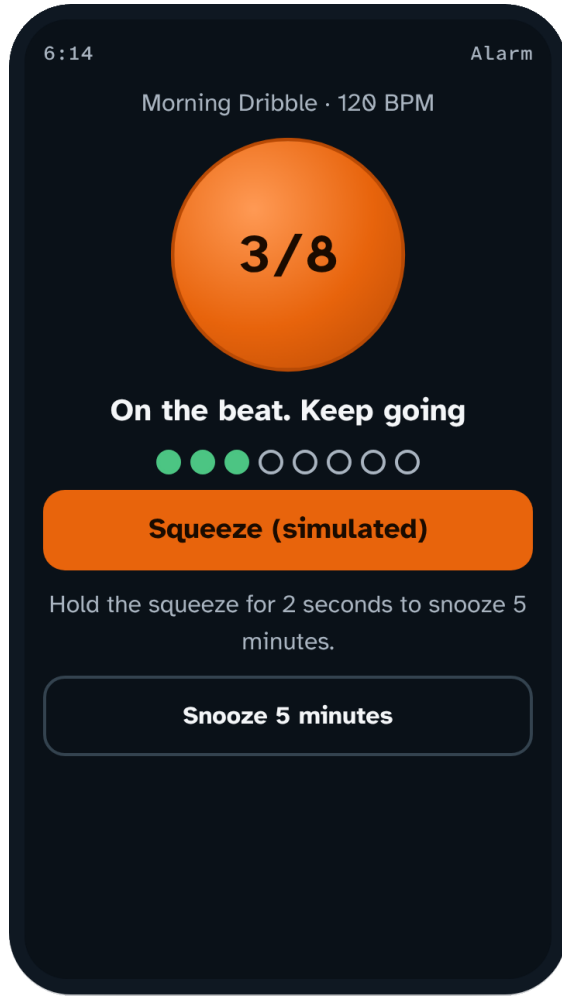
Fully charged

One soft beep, so nobody wakes up.

Ask any time

Settings > "Say battery level"

Squeeze on every BOOM to stop the alarm



One BOOM to follow

A loud drum hit with a bell on top, about once a second. The rest of the music stays quiet.

Only on-beat squeezes count

Keeping the rhythm proves you are awake. Choose 4, 8 or 12 squeezes.

Feel it as well as hear it

The ball buzzes and glows on every BOOM.

Hold to snooze

Hold the squeeze for 2 seconds to snooze for 5 minutes.